JUNE 26， 2026

RECOVERY PROGRESS REPORT

Week 2

This Week’s Tasks

- - XBEE Antenna Research
- - General purchases and beginning of X-BEE importation process
- -Free RTOS Configuration Review
- - 3D printing Shear Pins
- - Implement circuit changes for dual memory storage
- - Documentation
- - Code cleaning
- - cooking crimson powder

Extracting data from the flash memory

1. Fixing the flash memory erase

The initial software attempted to clear the SPI flash memory using a standard byte-overwrite method. This was ineffective because NOR flash hardware strictly requires a dedicated sector-erase command to reset memory bits and prevent data corruption.

2. Reading the data correctly

The recovery program was trying to read complex binary code from a file named 'flight', while the flight computer was actually saving simple plain text into a file named 'data'.

The flight computer was saving the telemetry data as simple, plain text . But the recovery program was trying to read it as complex binary code. Because of that mismatch, the recovered data just looked like gibberish.

G13 | 0.62

	A	B	C	D	E	F	G	H	I	J
1	record_number	operation_mode	state	ax	ay	az	pitch	roll	gx	gy
2	3456	0	0	0.01	0	0.97	0.42	-0.81	-10.79	3
3	3457	0	0	0.01	-0.01	0.96	0.9	-0.61	-11.25	6
4	3458	0	0	0.01	-0.01	0.96	0.67	-0.89	-11.55	
5	3459	0	0	0.01	-0.01	0.97	0.62	0.03	-10.34	4
6	3460	0	0	0.01	0	0.96	0	-0.2	-10.24	
7	3461	0	0	0.02	-0.01	0.98	0.25	-0.34	9.48	-27
8	3462	0	0	0.01	0	0.97	0.76	0.2	-10.52	6
9	3463	0	0	0.01	0	0.97	0.7	-0.29	-8.93	3
10	3464	0	0	0.03	0	0.96	1.76	0.57	-9.73	-4
11	3465	0	0	0.01	0	0.98	0	-0.43	-9.12	3
12	3466	0	0	0.02	-0.01	0.97	0.45	-0.8	-10.76	7
13	3467	0	0	0.01	0	0.97	0.62	-0.2	-6.55	-1
14	3468	0	0	0.01	0	0.96	0.45	-0.29	-15.67	13
15	3469	0	0	0.03	0	0.93	1.4	-0.33	-11.46	1
16	3470	0	0	0	0	0.97	0	-0.38	-12.04	3
17	3471	0	0	0.01	0	0.97	0.73	-0.35	-11.19	0
18	3472	0	0	0.03	0.01	0.95	1.71	0.03	-34.54	4
19	3473	0	0	0.02	0	0.95	1.43	0.06	-13.99	
20	3474	0	0	0	-0.01	0.98	-0.42	-0.88	-15.34	25
21	3475	0	0	0	-0.01	0.98	-0.17	-0.57	-5	-1
22	3476	0	0	0	-0.02	0.97	0.73	-0.75	-12.47	7
23	3477	0	0	0.01	0	0.96	0.42	-0.29	-9.57	0
24	3478	0	0	0.01	-0.01	0.96	1.2	-0.49	-10.4	

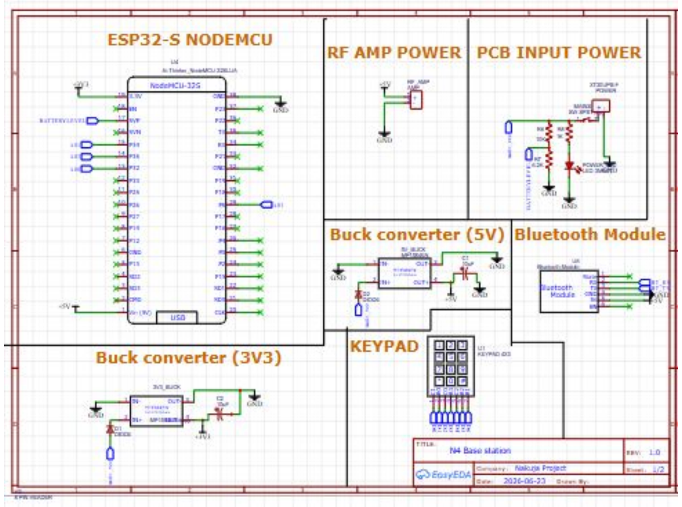
flight_data

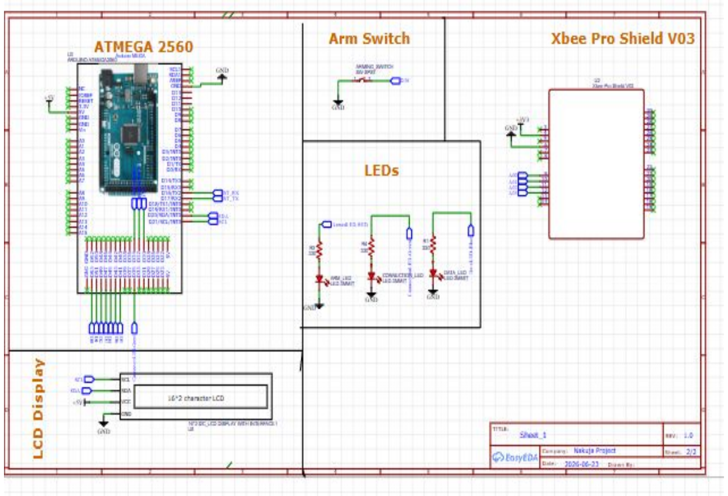
Ready | Accessibility: Unavailable | 100%

1. Successfully extracted the sensor data from the Flight Computer (FC)
2. Validation will be conducted accordingly
3. GPS not connected - TBD

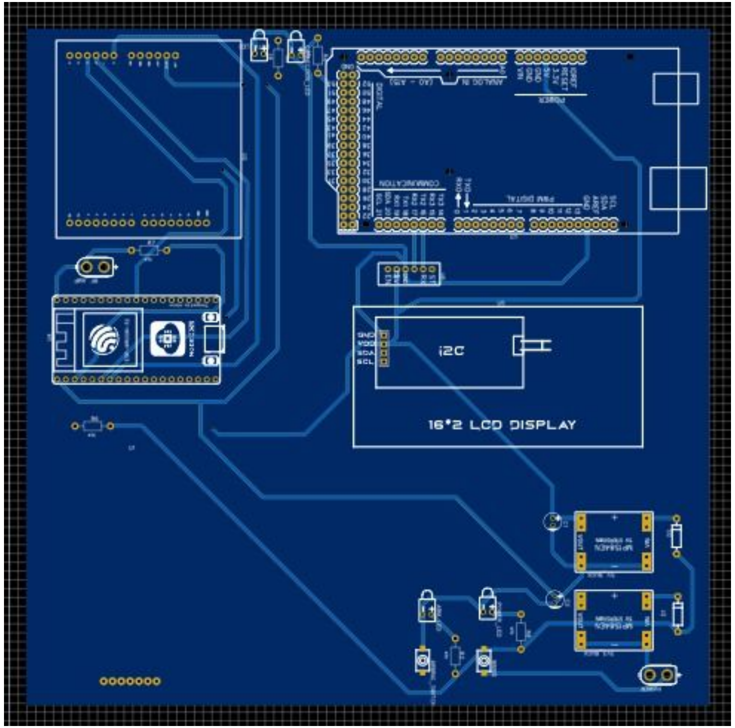
Updated the base station schematic and designed the PCB

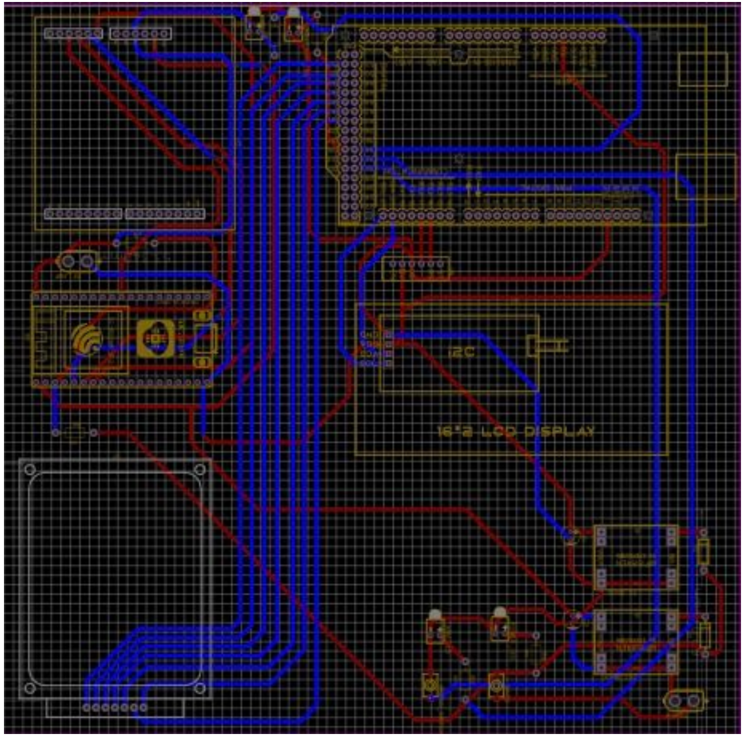
- No documentation hence we had to reverse engineer Glen’s model.
- Didn’t etch since we don’t have the copper clad.
- Route PCB





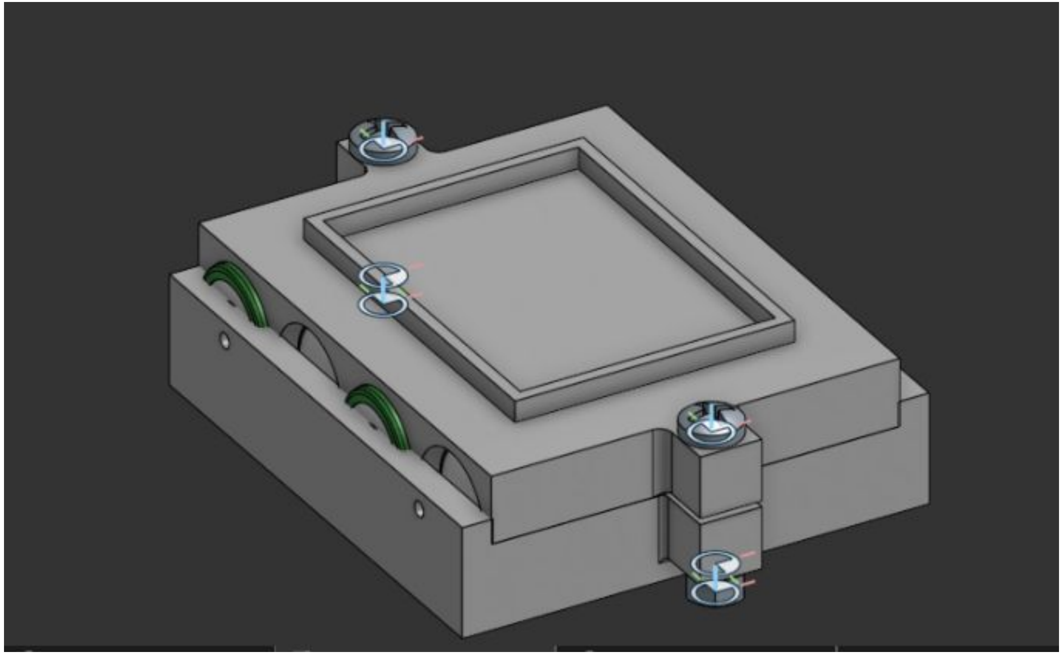
Routed PCB





Designed Battery Holder for Base Station

Is yet to be 3D printed.



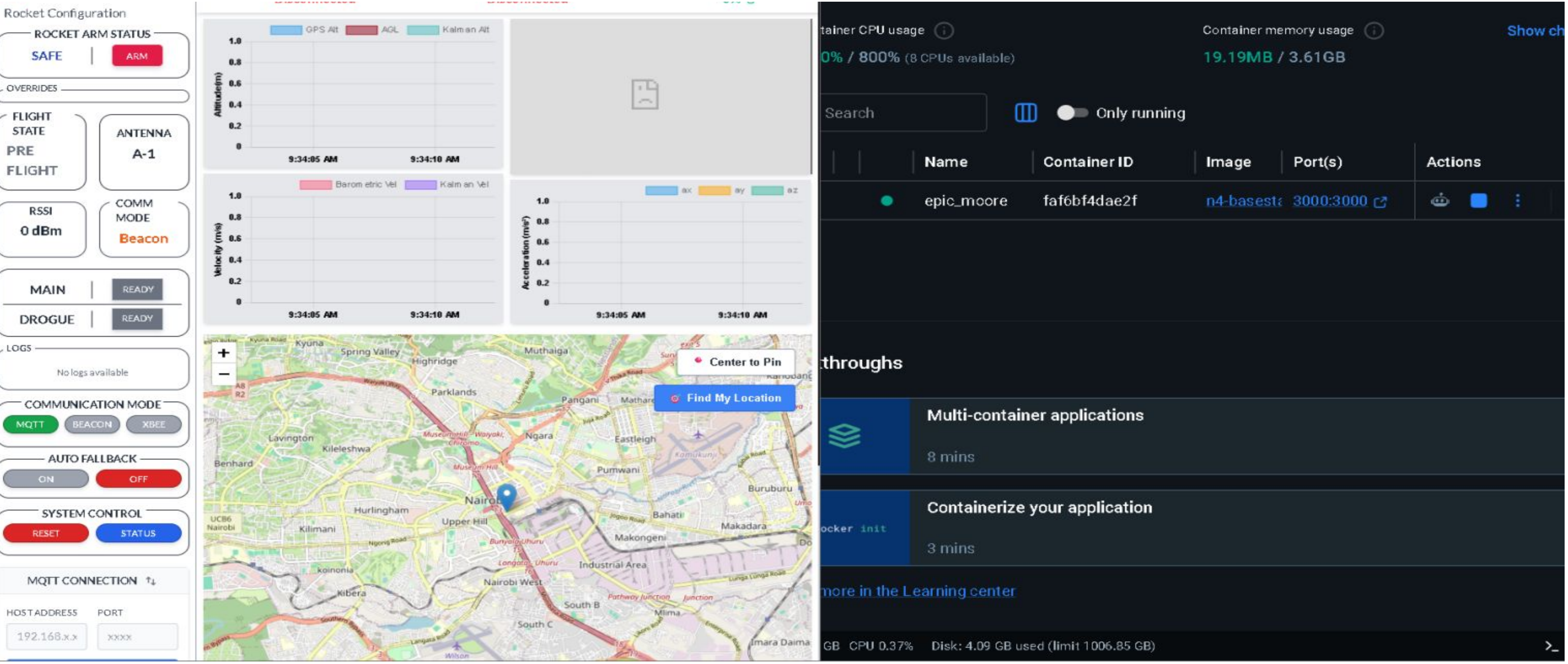
Code Review and Cleaning

- Finished code review in preparation for pop test.
- Began code cleaning: balancing tasks between the cores.
- Generated simulation data from OpenRocket and added functionality for loading data from the csv file.

```
# Simulation 3 (Up to date)
# 1243 data points written for 4 variables.
# Simulation warnings:
#   Recovery device deployment at high speed (30.4 m/s):  "Main Parachute"
#
# Time (s),Altitude (m),Vertical velocity (m/s),Vertical acceleration (m/s²)
# Event IGNITION occurred at t=0 seconds
# Event LAUNCH occurred at t=0 seconds
0,0,0,0
0.01,1.678e-4,0.067,19.864
0.02,0.002,0.414,49.539
0.03,0.009,1.058,79.234
# Event LIFTOFF occurred at t=0.04 seconds
0.04,0.024,1.851,79.546
0.05,0.046,2.648,79.857
0.06,0.077,3.449,80.169
0.07,0.115,4.252,80.478
0.08,0.162,5.058,80.787
0.09,0.217,5.868,81.096
0.1,0.279,6.68,81.403
0.11,0.35,7.496,81.711
0.12,0.429,8.314,82.018
0.13,0.517,9.136,82.323
0.14,0.612,9.961,82.628
0.15,0.716,10.788,82.933
0.159,0.819,11.559,83.213
0.168,0.922,12.28,83.475
# Event LAUNCHROD occurred at t=0.176 seconds
0.176,1.025,12.96,83.722
0.18,1.077,13.29,83.841
0.186,1.157,13.786,84.018
0.195,1.282,14.531,84.283
0.21,1.513,15.819,84.741
Flight Simulation from Open Rocket.csv [dos] (10:27 26/06/2026)
```

```
#if USE_SIMULATION
bool load_sim_file(const char *filename)
{
    File file = SPIFFS.open(filename);
    if (!file) {
        debugln("Failed to open simulation file!");
        return false;
    }

    sim_count = 0;

    while (file.available()) {
        String line = file.readStringUntil('\n');
        line.trim();

        if (line.length() == 0) continue; //skip blank lines
        if (line[0] == '#') continue; //skip comments
        if (!isdigit(line[0])) continue; //skip title

        float t, alt, vel, accel;

        if (sscanf(line.c_str(),
                   "%f,%f,%f,%f",
                   &t,
                   &alt,
                   &vel,
                   &accel) == 4) {
            if (sim_count < MAX_SIM_RECORDS) {
                sim_data[sim_count].time_s = t;
                sim_data[sim_count].altitude_m = alt;
                sim_data[sim_count].velocity_mps = vel;
                sim_data[sim_count].acceleration_mps2 = accel;
main.cpp[+] [dos] (16:02 25/06/2026)                                475,2-8 10%
```

Dockerized the Base Station



Use of COTS altimeter as backup

https://www.apogeerockets.com/Electronics-Payloads/Dual-Deployment/Blue-Raven-Altimeter

Next weeks tasks

- Etch Base-Station PCB
- Pop test
- 3D - Printing shear pins ,Battery Holder, explosion caps
- Kalman Filter Testing
- Make circuit changes for dual memory storage
- Verification of parachute design to ensure they are able to handle the new motor etc
- Changes to base-station Arduino MEGA code
- Extending parachute suspension lines